

Cybersecurity Portfolio

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Projects

1. Exploring & Navigating a Cloud-Hosted Cybersecurity Lab Environment

Overview

Explored a cloud-hosted enterprise lab environment composed of multiple Windows and Linux virtual machines. The focus was on understanding system roles, network segmentation, and tooling required for cybersecurity analysis.

Key Activities

- Navigated Windows Server and Linux environments
- Identified system roles (Domain Controller, servers, client workstations)
- Reviewed network segmentation and access paths
- Prepared systems for security analysis

Skills Demonstrated

- Enterprise network architecture awareness
- Windows Server & Linux navigation
- System role identification

- Security environment preparation

CySA+ Alignment

1.1, 2.5

2. Configuring Security Controls

Scenario

Configured and validated **preventive, detective, directive, and corrective controls** to improve the organization's security posture.

Activities

- Implemented firewall rules
- Configured Group Policy security settings
- Controlled service behavior and permissions
- Tested control effectiveness

Skills Demonstrated

- Security control categorization
- Hands-on server and policy configuration
- Control validation

CySA+ Alignment

1.1, 2.5

3. Reviewing Indicators of Compromise (IoC) & Threat Intelligence Sources

Scenario

Researched and applied multiple threat intelligence sources to strengthen detection and vulnerability awareness.

Activities

- Reviewed IP-, domain-, and hash-based IoCs
- Analyzed MITRE ATT&CK techniques
- Researched CVE repositories
- Compared tactical, operational, and strategic intelligence

Skills Demonstrated

- Threat intelligence sourcing
- IoC analysis
- Vulnerability baseline understanding

CySA+ Alignment

1.1, 1.4, 1.5, 2.5

4. Performing Threat Hunting

Scenario

Conducted proactive threat hunting across multiple systems using logs, network analysis, and DNS investigation.

Activities

- Identified suspicious IP traffic from firewall logs

- Used netstat to detect abnormal connections
- Investigated DNS anomalies linked to command-and-control activity
- Performed cross-system investigative triage

Skills Demonstrated

- Network and endpoint threat hunting
- IoC tracking (IP, DNS, ports, processes)
- Incident analysis

CySA+ Alignment

1.2, 1.3, 1.4, 3.2

5. Performing System Hardening

Scenario

Hardened Windows and Linux systems by removing unnecessary components and securing essential services through iterative improvement.

Activities

- Managed Windows device drivers
- Modified hosts file name resolution
- Removed unnecessary software and services
- Configured restrictive firewall rules
- Managed Linux file permissions

Skills Demonstrated

- System hardening methodology
- Attack surface reduction
- Linux permission management

CySA+ Alignment

1.1, 2.5

6. Configuring Automation for Security Operations

Scenario

Integrated threat intelligence feeds with automated security controls to improve response speed and operational efficiency.

Automation Tasks

- Automated firewall rule updates using malicious IP feeds
- Scripted malware removal using hash-based intelligence
- Implemented DNS blocking for malicious domains
- Scheduled automated remediation tasks

Skills Demonstrated

- Security automation scripting
- Threat intelligence integration
- Automated remediation workflows

CySA+ Alignment

1.1, 1.3, 1.4, 1.5, 2.1, 2.5

7. Vulnerability Assessment Using OpenVAS (Greenbone)

Overview

Performed a network vulnerability assessment using **OpenVAS (Greenbone Vulnerability Manager)** to identify security weaknesses and evaluate remediation strategies.

Environment

- Kali Linux (Virtual Machine)
- OpenVAS / Greenbone Vulnerability Manager
- Local network subnet

Methodology

- Installed and initialized OpenVAS and vulnerability feeds
- Verified system readiness and service status
- Created a vulnerability scan task using the Advanced Task Wizard
- Executed a comprehensive network scan
- Reviewed vulnerability findings, CVEs, and remediation guidance

Key Findings

- Identified informational and medium-severity vulnerabilities
- Detected misconfigurations and outdated services
- Evaluated risk exposure across network assets

Risk & Impact

Unpatched or misconfigured systems increase the likelihood of unauthorized access, service exploitation, and lateral movement.

Remediation Recommendations

- Apply system patches and updates
- Disable unnecessary services and open ports
- Strengthen firewall and network segmentation
- Perform regular vulnerability scans

Skills Demonstrated

- Vulnerability scanning and analysis
- OpenVAS / Greenbone configuration
- Risk assessment and remediation planning
- Security reporting

CySA+ Alignment

1.1, 2.5